## Prototype refinement

Iterative refinement

Support prototypes

Soft assignments

Weighted soft prototypes

Refined prototypes

Blend

Weighted mean

Unlabeled features

Retain support anchors

Sharpening

Fixed signed gates

Assignment sharpening

## Class prompts P

Node-wise correction

## Post-training: calibrated readout

Attribute similarity

Neighbor selection

Attribute graph

a

b

c

Centered propagated features

Teacher features from M1

Exclude self

Similarity weights

Prototype classifier

Initial probabilities

From Prompt scores s

Fixed W

## Shared frozen graph encoder

LeakyReLU

Input graph

Layer 1

Hidden-feature correction

Layer 2

Adapted representations

Local agreement → node temperature

Initial classes

Weighted by W

$a_i = \sum_j W_{ij}\mathbf{1}[c_j = c_i]$

$T_i = 1 + \lambda(1 - a_i)$

$F_{0,i} = \mathrm{softmax}(s_i/T_i)$

Prediction softening

## Prompt learning: direction and norm adaptation

Support loss

Shared gradient G

Rescaled direction $\tilde{D}$

Same ray; only the length changes

Adam reference A

Use its norm, not its direction

Shampoo direction D

Common origin

Dashed-circle radius = Adam norm

$\|\tilde{D}\|_F = \|A\|_F$

Match the whole-matrix norm

$P \leftarrow P - \eta\tilde{D}$

Parameters move in the negative direction

Current probabilities

Neighbor propagation

Updated probabilities

Iterate

Calibrated $F_0$

Fuse

Weighted neighbor probabilities

Updated class probabilities

Clamp support labels

Predicted classes

Frozen backbone

Support node

Only P is updated